

Dear Luc,

Here is a global proof of the existence of a DR complex giving rise to mixed Hodge theory.

Objects considered. The derived category of the following category of bifiltered complexes (K^*, d, W, F) :

$$\begin{aligned} K^i & : \text{ a quasi-coherent sheaf, as are } W_j K^i \text{ and } F^j K^i, \\ d & : \text{ a first-order differential operator, respecting } W \text{ and } F, \\ \text{Gr}_F(d) & : \mathcal{O}\text{-linear.} \end{aligned}$$

Variants.

- a) Require only that $\mathcal{H}^i \text{Gr}_F K$ be quasi-coherent.
- b) In this letter we restrict ourselves to F , rather than to W .

Remarks.

- a) There are functors Rf_* .
- b) For $X_\bullet \xrightarrow{\varepsilon} X$, define $R\varepsilon_*$ as follows. After resolving and applying ε_* , one obtains a double complex K^{ij} on X , with W, F , where i is the index of X_i and j is the degree. One takes the associated simple complex; for F , the simple complex associated to the F^i , and, for W , the diagonal between i and W :

$$W^i = \bigoplus_{i'+i''=i} W^{i'} K^{i'',*}.$$

- c) There is inverse image by a smooth morphism $f : Y \rightarrow X$:

$$f^* K :=_{\text{dfn}} \Omega_Y^* \otimes_{f^{-1}\Omega_X^*} f^{-1} K.$$

What one wants. Construct functorially, for $U \subset X$, with X proper, such an object on X , whose $R\Gamma$ “is” a mixed Hodge complex and gives rise to the mixed Hodge structure on $H(U, \mathbb{C})$. In particular, the assertions of 8.19 in Hodge III are true.

I realize that asking for this is too much: only (K^*, d, F) can be canonical in this way. Why?

The construction is given in Hodge III, 8.3: take a resolution

$$(U_\bullet, X_\bullet) \xrightarrow{\varepsilon} (U, X)$$

as in Hodge III, and, for $D_\bullet = X_\bullet - U_\bullet$,

$$R\varepsilon_*(\Omega_{X_\bullet}^*(\log D_\bullet), W, F).$$

The problem is the independence of the choice of $U_\bullet, X_\bullet$.

Thus let

$$\begin{array}{ccc} (U'_\bullet, X'_\bullet) & \xrightarrow{\varphi} & (U_\bullet, X_\bullet) \\ \downarrow \varepsilon' & & \downarrow \varepsilon \\ (U, X) & & (U, X) \end{array}$$

and

$$K = R\varepsilon_*\Omega(\log), \quad K' = R\varepsilon'_*\Omega(\log), \quad \varphi^* : K \longrightarrow K'$$

on X , proper. We have:

- 1) $R\Gamma(X, K) \xrightarrow{\sim} R\Gamma(X, K')$. This is $H^*(U, \mathbb{C})$.
- 2) $R\Gamma(X, \mathrm{Gr}_F^i K) \xrightarrow{\sim} R\Gamma(X, \mathrm{Gr}_F^i K')$. This is $\mathrm{Gr}_F^i H^*(U, \mathbb{C})$, since, for F , $E_1 = E_\infty$.

Now one uses the method of SGA 4 $\frac{1}{2}$, finiteness. Proceeding by induction on $\dim X$, and noting that what is true over $\mathbb{C}$ is also true over any field of characteristic 0, one finds that

$$\mathrm{Gr}_F^i(K) \longrightarrow \mathrm{Gr}_F^i(K')$$

is a quasi-isomorphism away from finitely many points, rather away from a skyscraper, and that the third vertex of the triangle has zero cohomology. Restricting to U , it is a quasi-isomorphism everywhere.

Conclusion. For $U/\mathbb{C}$, there is a complex (K^*, d, F) , functorial up to quasi-isomorphism, with

$$\begin{cases} \mathcal{H}(K^{*,\mathrm{an}}) = \mathbb{C}_U, & \mathbb{H}(U, K^*) = \mathbb{H}(U^{\mathrm{an}}, K^{*,\mathrm{an}}) = H(U, \mathbb{C}), \\ \text{for } U \text{ proper, the spectral sequence of } F \text{ degenerates at } E_1, & \text{and abuts to the Hodge filtration.} \end{cases}$$

Yours,
P. Deligne